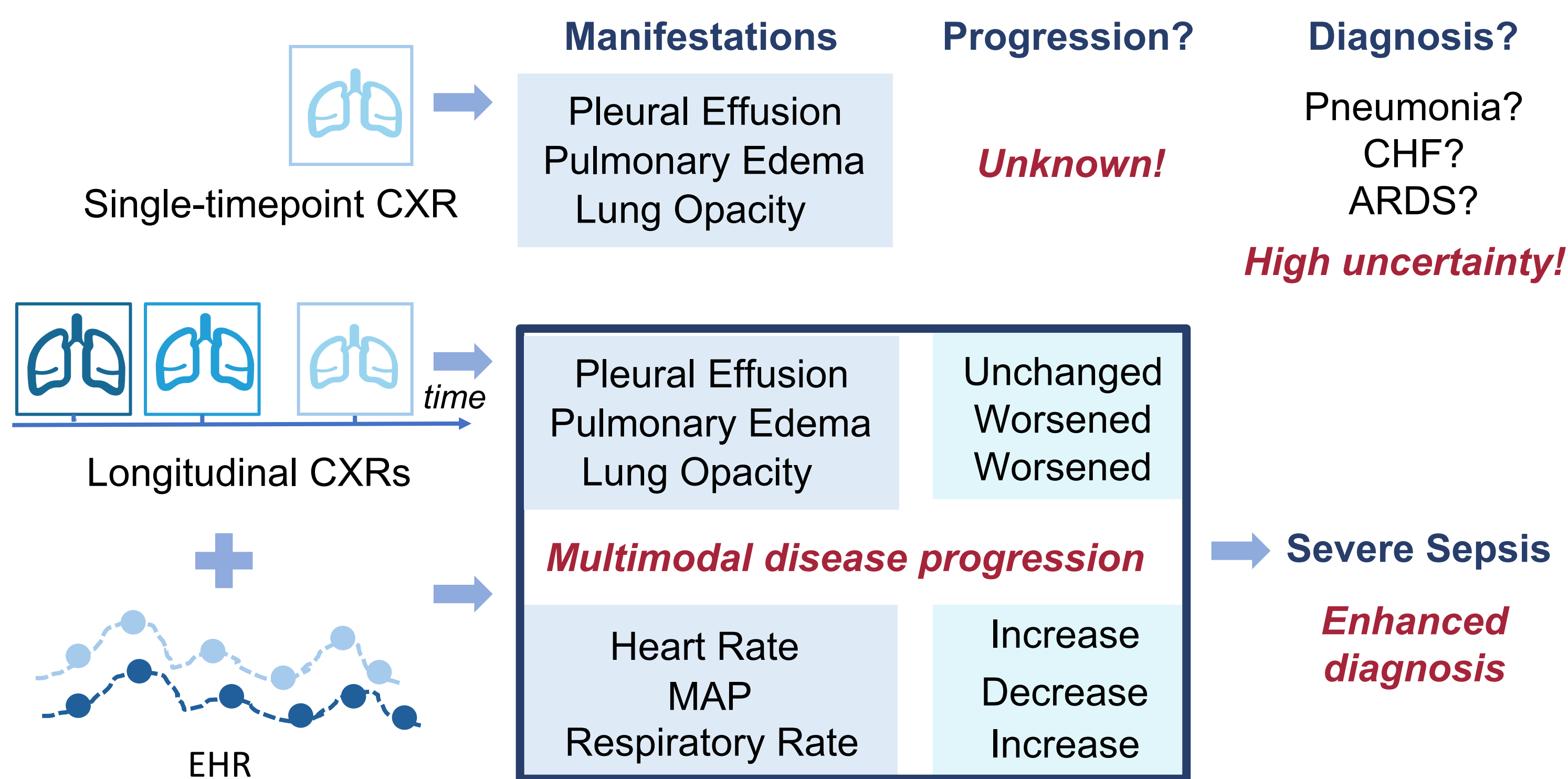


Multimodal Disease Progression Modeling via Spatiotemporal Disentanglement and Multiscale Alignment

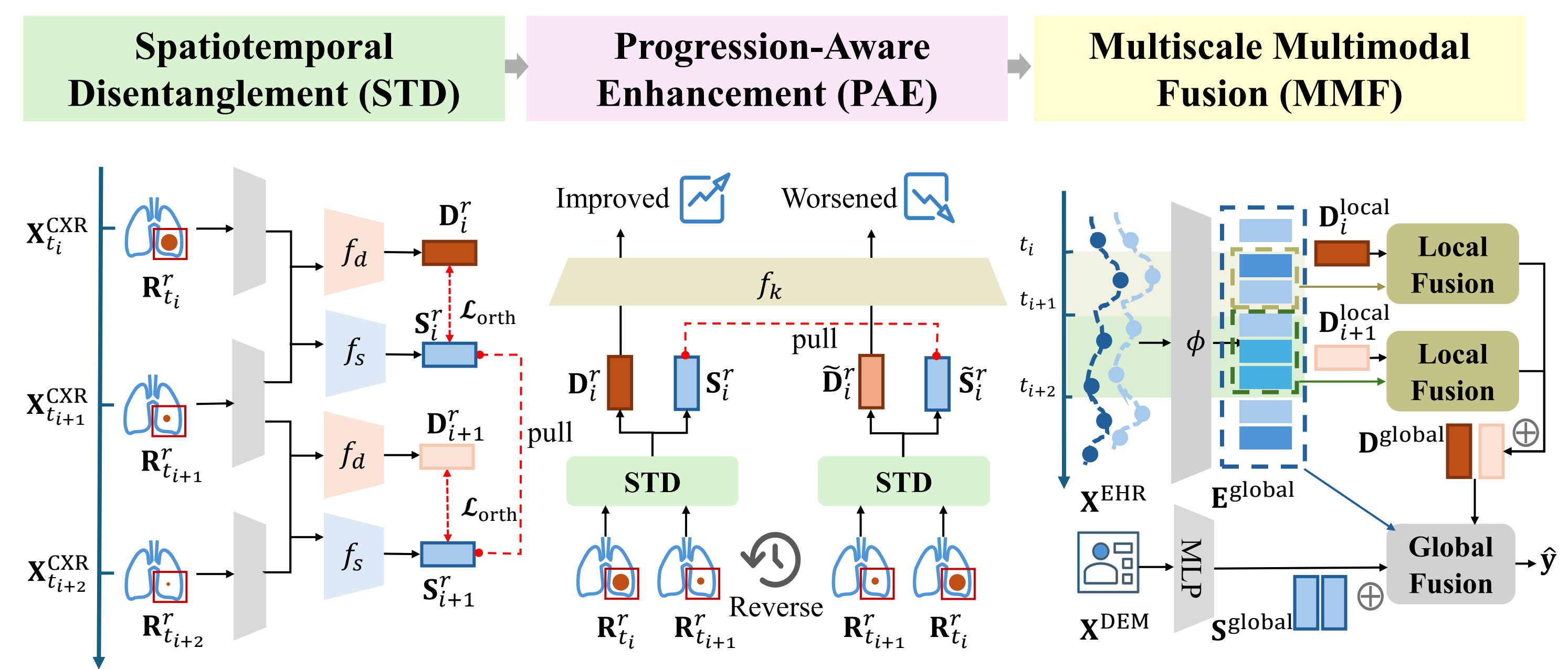
Chen Liu[▲], Wenfang Yao[▲], Kejing Yin[◆], William K. Cheung[◆], Jing Qin[▲]

Motivation

A real case from MIMIC:



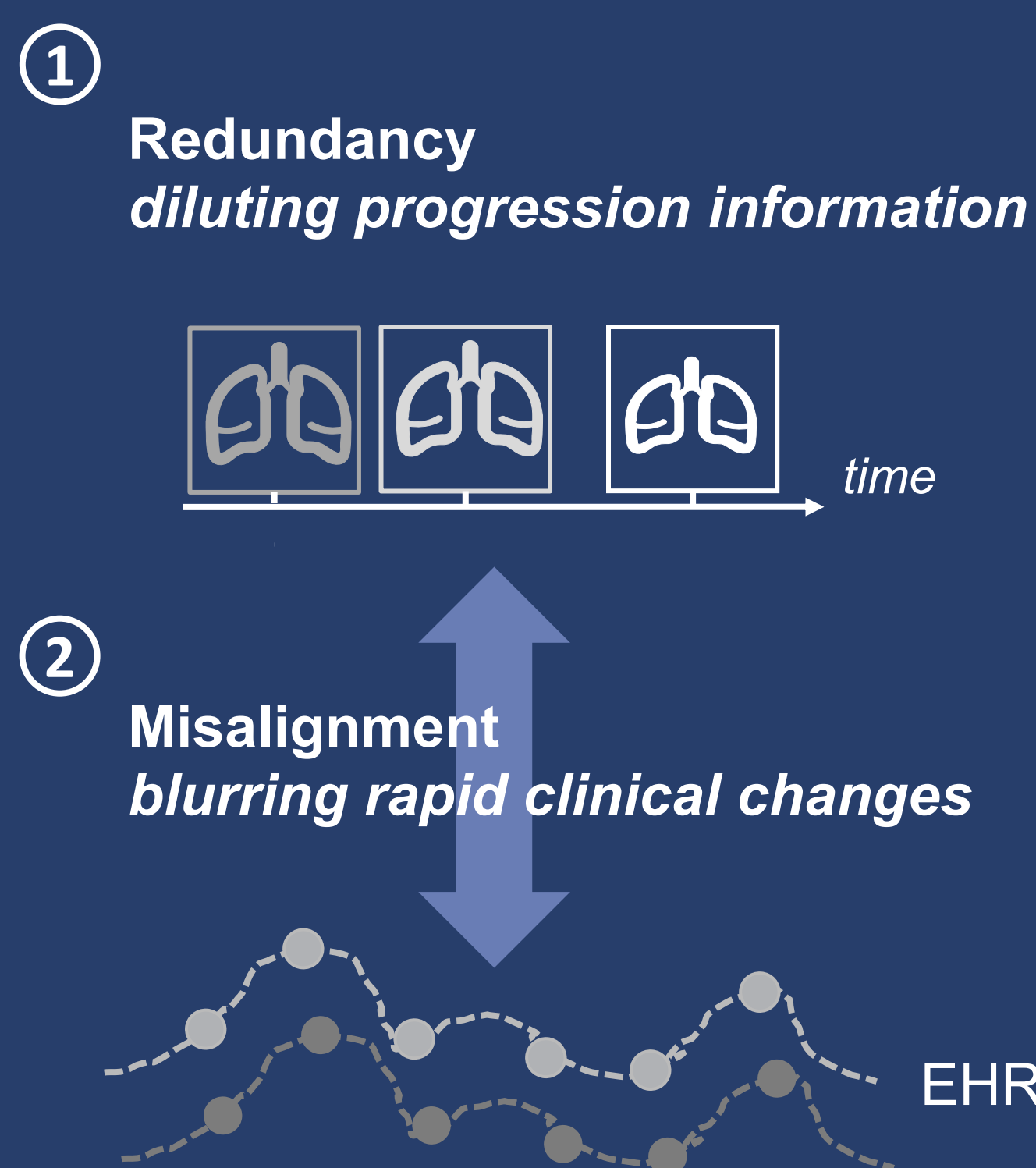
Method: DiPro



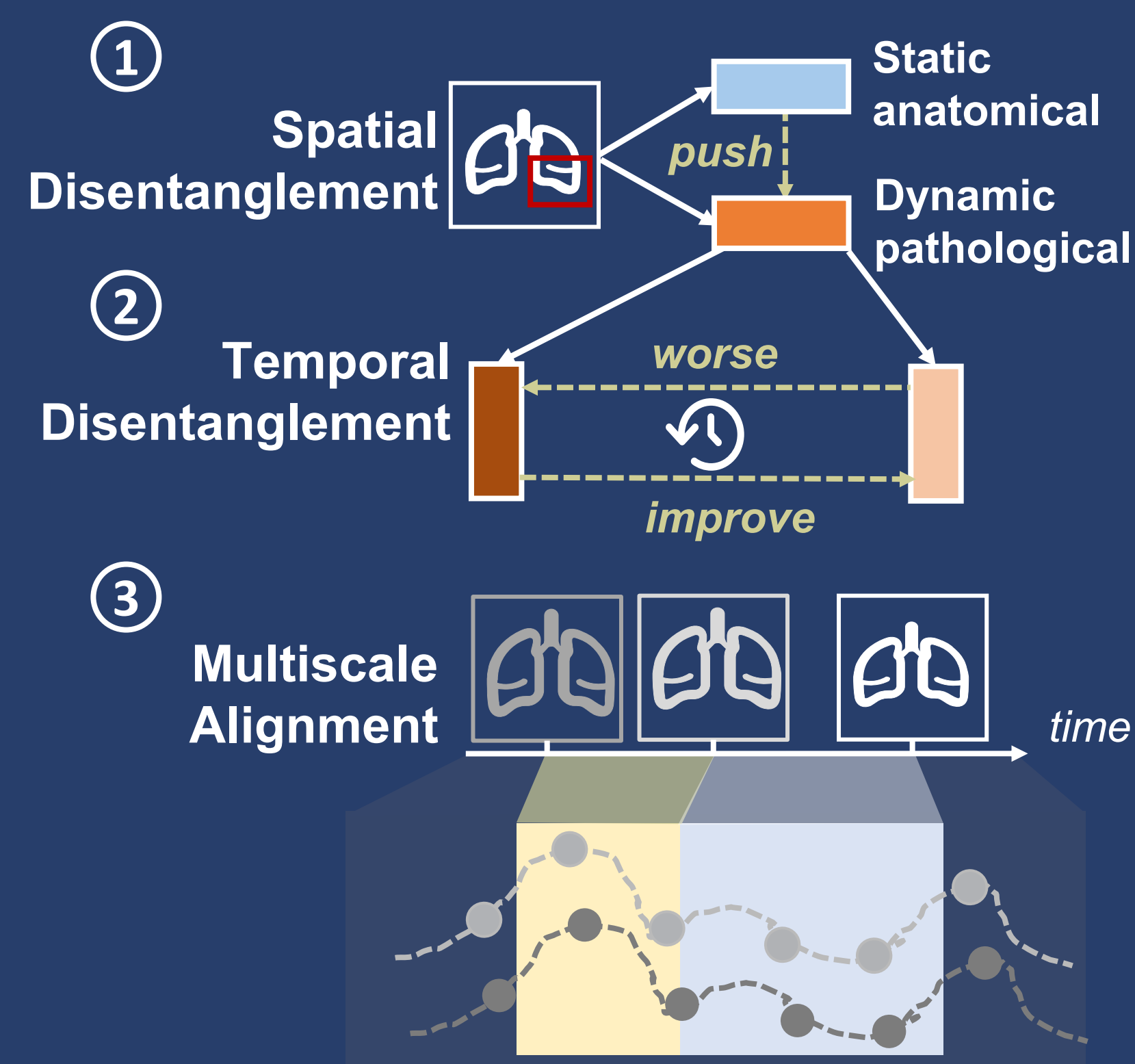
TL; DR.

DiPro disentangles static anatomy and dynamic pathology from *longitudinal CXRs* and aligns them with *EHRs* across timescales for precise **disease progression modeling**.

Challenges



Our Solutions



Key Insights:

- Disentangle Dynamic from Static Representations
- Incorporate Progression-Direction Awareness
- Multiscale Fusion of Longitudinal Multimodal Data



Results

Disease Progression Identification

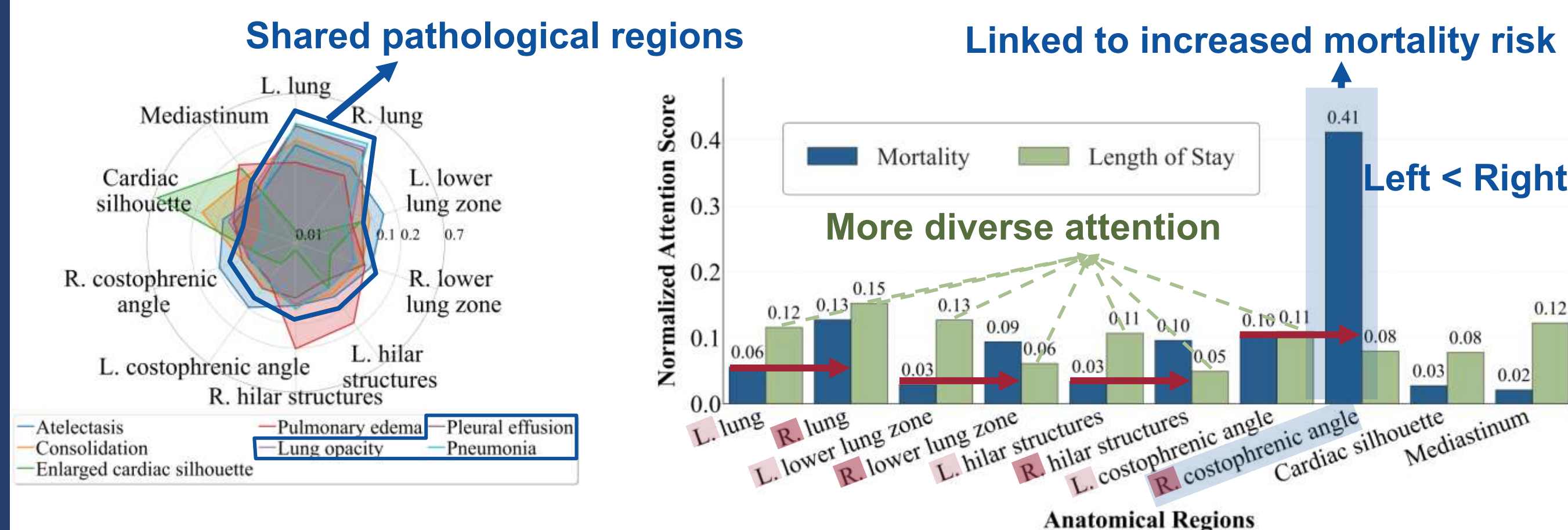
Method	Precision	Recall	F1	AUPRC	AUROC
Unimodal Methods (CXR)					
CheXRelNet [14]	0.395±0.015	0.392±0.010	0.389±0.010	0.394±0.010	0.574±0.011
CheXRelFormer [33]	0.389±0.044	0.379±0.033	0.354±0.032	0.372±0.023	0.551±0.041
SDPL [13]	0.408±0.006	0.406±0.020	0.393±0.010	0.417±0.032	0.609±0.031
DiPro (ours)	0.475±0.004	0.452±0.011	0.453±0.009	0.468±0.013	0.651±0.016

Multimodal Methods					
UTDE [19]	0.481±0.017	0.462±0.002	0.449±0.005	0.472±0.014	0.659±0.011
UMSE [20]	0.353±0.011	0.361±0.009	0.352±0.013	0.364±0.006	0.544±0.004
MedFuse [17]	0.423±0.049	0.413±0.045	0.409±0.042	0.422±0.040	0.530±0.030
DrFuse [18]	0.442±0.009	0.461±0.007	0.429±0.010	0.438±0.003	0.628±0.002
DiPro (ours)	0.484±0.008	0.471±0.024	0.466±0.018	0.478±0.018	0.664±0.013

General ICU Prediction

Method	CXR Used		Mortality		Length of Stay	
	Last	Long.	AUPRC	AUROC	Kappa	ACC
UTDE [19]	✓		0.717±0.019	0.887±0.004	0.160±0.016	0.381±0.013
UMSE [20]	✓	✓	0.710±0.019	0.887±0.012	0.195±0.031	0.400±0.021
		✓	0.722±0.039	0.896±0.012	0.217±0.013	0.419±0.010
MedFuse [17]	✓		0.686±0.018	0.869±0.011	0.213±0.012	0.413±0.004
		✓	0.716±0.018	0.881±0.005	0.210±0.039	0.412±0.027
DrFuse [18]	✓		0.709±0.012	0.865±0.014	0.114±0.048	0.338±0.041
		✓	0.684±0.008	0.854±0.017	0.142±0.014	0.360±0.011
DiPro (Ours)			0.712±0.009	0.885±0.003	0.226±0.019	0.427±0.014
	✓		0.742±0.003	0.897±0.002	0.248±0.008	0.440±0.007

Averaged attention weights of CXR regions in different tasks



(a) Disease Progression Identification

(b) General ICU Prediction